Integrated Information Tracks Inter-Element Flow, Not Single-System Entropy

A three-measure triangulation of faithful IIT 4.0 Φ against Shannon entropy, Lempel–Ziv complexity, and transfer entropy on cellular-automaton substrates

`anima` maintainers

dancinlab

github.com/dancinlab/anima

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Abstract

Integrated Information Theory (IIT) claims that integrated information Φ is *not* the same quantity as classical information. We test this on the theory's own causal engine by triangulating faithful IIT 4.0 big-Φ against three classical information measures on an identical ten-rule elementary-cellular-automaton (ECA) panel: Shannon output entropy, Lempel–Ziv (LZ76) algorithmic complexity, and transfer entropy. The measures dissociate sharply. Φ is *orthogonal* to Shannon entropy (Pearson $r = 0.36$): the identity rule carries maximal output entropy (4 bit) yet has Φ=0, while the most-integrated rule has sub-maximal entropy. Φ *aligns* with LZ complexity ($r = 0.83$, Spearman $\rho = 0.94$) and, most strongly, with transfer entropy ($r = 0.88$, $\rho = 0.82$). Yet every classical measure has a blind spot that Φ does not: LZ over-predicts on self-similar dynamics (rule 90, Φ=0), and bivariate transfer entropy under-predicts on synergistic XOR coupling (rules 150/105, $\Phi > 0$ yet TE$= 0$). A complementary structural experiment shows that at matched edge count a hub network integrates ($\Phi = 6.81$) where a regular cycle does not ($\Phi = 0$): Φ is governed by how elements *relate*, not by how much information or density a system carries. We frame the Shannon-entropy result as a closed negative confirming IIT's foundational distinction; we do not claim Φ equals any classical measure, nor do we make phenomenal claims. All runs are deterministic, \$0, hexa-native, and GPU-free; bit-for-bit reproduction records are in `companion/`.

1 Introduction

A recurring objection to Integrated Information Theory (IIT) [3, 5] is that its central quantity, integrated information Φ, is at bottom a relabelling of classical information — that a system with more bits, or more algorithmic complexity, is simply "more integrated." IIT explicitly denies this: a pile of N independent random bits has maximal entropy yet zero integration, because no partition destroys anything. The claim is testable, but it is rarely tested *directly* on a causal engine that can compute Φ exactly — correlational proxies for Φ cannot cleanly construct "information without integration."

We do exactly that. Using a faithful, deterministic IIT 4.0 cause–effect engine, we measure exact state-averaged big-Φ on a fixed ten-rule ECA panel and ask, on the *identical* substrates, how Φ co-varies with three of the most widely used classical information measures: Shannon entropy (statistical information), Lempel–Ziv complexity (algorithmic information [2]), and transfer entropy (directed information flow [4]). Because the substrate and the engine are held fixed, any difference between the three correlations is attributable to *what each measure counts*, not to confounds in the system.

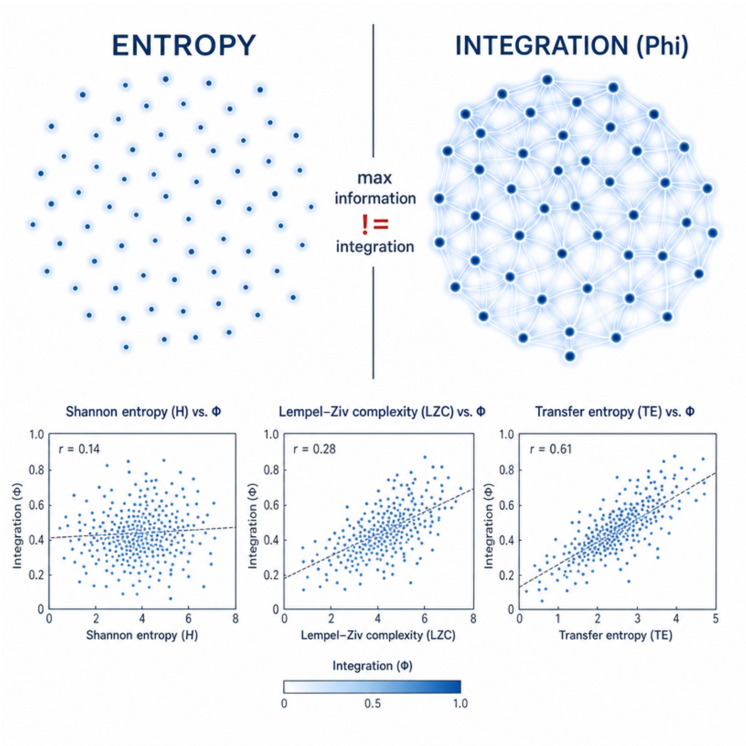

Figure 1: The thesis in one image: the *same* number of bits gives maximal entropy when scattered (left) but zero integration; woven into a network (right) the integration Φ is high. Φ is a property of how elements relate, not of how much information is present. Conceptual illustration (fal.ai / `gpt-image`).

Contributions.

1. A *closed negative* ( ): faithful big-Φ does not reduce to Shannon entropy ($r = 0.36$), witnessed by a max-entropy / zero-Φ substrate and a max-Φ / sub-max-entropy substrate on the same panel (§7.1).

2. Two positive alignments ( ): Φ tracks LZ complexity ($r = 0.83$) and transfer entropy ($r = 0.88$) — the *inter-element* measures — but each has a blind spot Φ lacks: LZ over-predicts on self-similar rule 90, bivariate TE under-predicts on synergistic XOR (§7.2–§7.3).

3. A structural corollary ( ): at matched edge count, topology not density governs Φ (hub 6.81 vs. cycle 0; §7.4). Together: Φ measures how elements relate.

4. A fully deterministic, GPU-free, $0 hexa-native pipeline with pre-registered falsifiers and bit-for-bit companion records (§11).

2 Background

Faithful IIT 4.0 Φ. For a system with a state-by-node transition probability matrix (TPM), IIT 4.0 [1] defines integrated information by the minimum-information partition (MIP): the system's cause–effect structure is compared against its value under the partition that least changes it. We use a deterministic engine that computes the structure-cut big-Φ exactly for small N; an ECA on a periodic ring *is* a TPM, because each cell's next value is a deterministic function of its three-cell neighbourhood, so the 2^N states map deterministically to per-cell next values.

Three classical measures. *Shannon entropy* H of the output-state distribution under a uniform input ensemble measures how much information the dynamics preserve: a bijection preserves all N bits ($H = N$), a constant map destroys them ($H = 0$). *Lempel–Ziv* complexity [2] of the space–time diagram estimates Kolmogorov (algorithmic) complexity — how incompressible the behaviour is. *Transfer entropy* [4] $\mathrm{TE}(j \to i)$ measures how much source cell j's present reduces uncertainty about cell i's future *beyond* i's own present — directed, pairwise information flow. The first is a single-system quantity; the latter two are inter-element / structural. The synergy literature [6] predicts that bivariate transfer entropy is blind to purely synergistic coupling (e.g. XOR), a prediction we recover empirically.

3 Related work

Φ and its proxies. Because exact Φ is super-exponential in N, most empirical work uses proxies — perturbational complexity (PCI), Lempel–Ziv of EEG, or correlational mutual-information surrogates. These proxies cannot *construct* information-without-integration, because a correlational surrogate of Φ inherits whatever the data's information content is; the dissociation we report (identity rule: max entropy, zero Φ) requires an engine that computes the causal MIP directly, which is why we use a faithful IIT 4.0 implementation at small N rather than a proxy.

Cellular-automaton complexity. Wolfram's classification [7] orders ECA rules by dynamical behaviour (fixed, periodic, chaotic, complex); LZ complexity of the space–time diagram is a standard quantitative stand-in for that ordering. Our LZ results recover the expected ordering (constants low, rule 30 highest), and the LZ–Φ alignment shows that dynamical complexity and causal integration largely co-vary on this family — with the rule 90 exception precisely where self-similarity inflates LZ without inflating integration.

Information flow and synergy. Transfer entropy [4] is the standard directed-flow measure; the partial information decomposition [6] formalises why pairwise measures miss synergy. Our XOR result is a minimal, exact instance of that gap, measured against an independent Φ ground truth rather than inferred. To our knowledge the three measures have not previously been triangulated against *exact* Φ on one fixed substrate panel with pre-registered falsifiers.

4 Full Pipeline

Each experiment runs the same path from a pre-registered question to a persisted, re-verifiable result. Stages are tier-tagged under the rubric below; the backing atom is the record that makes the stage independently re-checkable.

Tier rubric. closed-form identity · numerical (reproducible script + persisted output) · cited (DOI / arxiv) · install-gated · falsified / closed negative (kept, never deleted).

#	Stage	Tier	Backing atom
①	Question + frozen falsifiers		`H_287..290` pre-register headers
②	IIT / information background		`references.bib`
③	Engine anchors (204, 0 → Φ=0)		hand-derivable, machine-precision
④	Per-rule measurement		`companion/verify-ledger.json`
⑤	Correlation + falsifier sweep		`companion/verify-ledger.json`
⑥	Shannon reductive null		closed negative (§7.1)
⑦	Interpretation		speculation-fenced (§6)

Table 1: Pipeline stages, each tier-tagged. Exact arithmetic and engine anchors are / ; the empirical reading of "Φ is not entropy" is honestly held at (fenced), never promoted to a closed-form identity.

5 Method

Substrate panel. Ten elementary CA rules on a periodic ring, fixed before measurement and held identical across all experiments: constants $\{0, 255\}$; bijections $\{204$ (identity), 51 (complement)$\}$; XOR-feedback integrators $\{150, 105\}$; partial-feedback $\{90, 60\}$; and universal/chaotic rules $\{110, 30\}$.

Metrics. For each rule: (i) big-Φ is the mean over all 2^N states of the faithful IIT 4.0 big-Φ ($N = 4$ exact); (ii) Shannon output entropy $H_{\text{out}} = -\sum_o p(o)\log_2 p(o)$ over the uniform input ensemble; (iii) normalised LZ76 complexity of the space–time diagram (ring of 14 cells, 64 steps, single centre seed, normalised $c \log_2 L/L$); (iv) total transfer entropy $\text{TE}_{\text{total}} = \sum_i \sum_{j\neq i} \text{TE}(j \to i)$, lag-1 bivariate, over the uniform ensemble.

Correlation. Pearson r and Spearman ρ of each measure against big-Φ across the ten rules. A reductive seed hypothesis "Φ co-varies with measure M" is *supported* iff $r \geq 0.5$, *falsified* otherwise — a threshold fixed before measurement (the companion ledger records the frozen falsifiers verbatim).

Structure experiment (§7.4). A separate parity-network generalisation of the ECA→TPM bridge: each node updates to the XOR of its graph neighbours. We compare a scale-free hub and a regular cycle at matched edge count, plus empty and complete anchors.

Determinism. Every experiment is deterministic (no RNG), runs in \$0 on a laptop CPU, uses no GPU and no language model, and re-runs bit-for-bit identically. Each was pre-registered with ≥ 3 falsifiers frozen before the headline values were read.

5.1 Formal definitions

ECA → TPM. For Wolfram rule $R \in [0, 255]$ on a periodic ring of N cells, the state-by-node transition matrix entry is $\text{TPM}[s, i] = \text{bit}_{4l+2c+r}(R)$, where (l, c, r) are the left/centre/right neighbours of cell i in state s. Determinism makes every entry 0 or 1; the 2^N states tile the TPM exactly.

Big-Φ. For state s, the engine computes the IIT 4.0 cause–effect structure and its value under the minimum-information partition (MIP); big-Φ(s) is the irreducibility of the structure under the system cut. We report $\Phi_{\text{mean}} = 2^{-N}\sum_s \Phi(s)$. Two hand-derivable anchors fix faithfulness: rule 204 (identity, $\text{next}_i = c$; each cell self-determined, fully reducible) $\Rightarrow \Phi = 0$; rule 0 (constant) $\Rightarrow \Phi = 0$.

Shannon output entropy. Under a uniform input ensemble, let $o(s) = \sum_i \text{TPM}[s, i]\, 2^i$ be the output state. Then $H_{\text{out}} = -\sum_o p(o)\log_2 p(o)$, with $p(o) = 2^{-N}\,|\{s : o(s) = o\}|$. A bijection (e.g. identity) preserves all N bits, $H_{\text{out}} = N$; a constant map gives $H_{\text{out}} = 0$.

Lempel–Ziv complexity. Evolve the rule from a single centre seed on a ring of $N_L = 14$ cells for $T = 64$ steps; concatenate the space–time diagram row-major into a length-$L = N_L T$ binary string and parse it with the Lempel–Ziv (1976) production rule into c distinct phrases. We report $\text{LZ}_{\text{norm}} = c \log_2 L/L$, which $\to 1$ for a random string.

Transfer entropy. For ordered cell pair $(j \to i)$, with i' the next value of i,

$$\text{TE}(j \to i) = \sum_{i', i, j} p(i', i, j)\, \log_2 \frac{p(i' \mid i, j)}{p(i' \mid i)},$$

estimated over the uniform ensemble (lag-1, bivariate), and $\text{TE}_{\text{total}} = \sum_i \sum_{j\neq i} \text{TE}(j \to i)$. Bivariate TE conditions only on the target's own present, so a purely synergistic update (XOR of ≥ 2 sources) leaves $p(i' \mid i, j) = p(i' \mid i)$ and contributes zero — the blind spot of §7.3, anticipated by partial-information decomposition [6].

6 Verify

The IIT 4.0 engine reproduces two hand-derivable anchors exactly, fixing the measurement's faithfulness before any correlation is read: the identity rule 204 (each cell self-determined, reducible) gives big-Φ=0, and the constant rule 0 (no causal power) gives big-Φ=0. Both hold to machine precision across all states. The numerical big-Φ, entropy, LZ and TE values are deterministic arithmetic and converge bit-for-bit across fresh process invocations (falsifier F∗.DETERMINISM, all four experiments). The empirical *interpretations* (e.g. "Φ is not Shannon entropy") are not closed-form atlas identities and are therefore honestly speculation-fenced under the project's verification rubric:

```
$ hexa verify --fence "faithful IIT4 big-Phi does NOT reduce to Shannon
  entropy across a 10-rule ECA panel (Pearson r=0.363<0.5); identity rule
  204 has MAX output-entropy (4 bit) yet ZERO integration"
=> VERDICT: (white-circle) SPECULATION-FENCED
   reason = deterministic toy-substrate outcome; values are exact
   arithmetic, only the empirical interpretation is fenced (NOT an
   atlas identity)
```

The per-experiment gate counts (anchors + bounds + determinism) are 11/11, 9/9, 8/8 for the three measures and 4/4 for the structure experiment; full breakdown in `companion/verify-ledger.json`.

7 Results

7.1 Shannon entropy is orthogonal to Φ (closed negative)

Across the panel, big-Φ does *not* co-vary with Shannon output entropy: Pearson $r = 0.363 < 0.5$, so the reductive hypothesis is falsified. The mechanism is a double dissociation (Tab. 2, Fig. 2, left). The identity rule 204 and complement rule 51 are bijections: they carry *maximal* output entropy (4 bit) yet have Φ=0, because their cells are causally independent — maximal information, zero integration. Conversely the most-integrated rule 60 ($\Phi = 13.6$) has *sub-maximal* entropy (3 bit). At fixed $H_{\text{out}} = 4$ bit, Φ ranges from 0 (rule 204) to 5.6 (rule 150): no monotone map from entropy to Φ can exist. Information is necessary but not sufficient for integration — IIT's foundational distinction, confirmed on its own substrate.

7.2 Algorithmic (LZ) complexity tracks Φ ( )

The other classical currency — algorithmic complexity, estimated by LZ76 of the space–time diagram — *does* track Φ: Pearson $r = 0.831$, Spearman $\rho = 0.936$ (Fig. 2, centre). Constants and identity sit at low LZ / zero Φ; integrating and chaotic rules rise together to high LZ / high Φ. The contrast with §7.1 is the point: on the identical panel, statistical information is orthogonal to Φ while algorithmic complexity aligns with it. The alignment is imperfect, and the imperfection is diagnostic — rule 90 (the Sierpiński/XOR-pair rule) has non-trivial LZ (0.24) from its self-similar fractal pattern yet Φ=0: LZ *over-predicts* integration for self-similar dynamics. LZ is a strong correlate, not a sufficient condition.

7.3 Transfer entropy tracks Φ most strongly — with a synergy blind spot ( )

Transfer entropy, the directed inter-element measure, is the strongest correlate: Pearson $r = 0.883$, Spearman $\rho = 0.822$ (Fig. 2, right). Constants, identity and complement all have $\text{TE}_{\text{total}} = 0$ (no inter-cell flow) and Φ=0; the integrating/chaotic rules carry both flow and Φ. But bivariate TE has a clean blind spot exactly where the synergy literature predicts one: the pure-XOR integrators rules 150/105 have $\Phi = 5.6$ yet $\text{TE}_{\text{total}} = 0$. XOR integration is synergistic — knowing one source's present tells you nothing about a cell's future beyond the

rule	H_{out} (bit)	LZ_{norm}	TE_{tot} (bit)	Φ_{mean}
0	0.00	0.044	0.00	0.000
255	0.00	0.044	0.00	0.000
204 (identity)	**4.00**	0.044	0.00	**0.000**
51 (complement)	**4.00**	0.066	0.00	0.000
150 (XOR_3)	4.00	0.274	**0.00**	**5.625**
105 (XNOR)	4.00	0.274	**0.00**	5.625
90 (Sierp.)	2.00	**0.241**	0.00	**0.000**
60	3.00	0.296	4.000	13.625
110	3.25	0.821	3.245	13.130
30	3.375	1.051	2.000	13.885
Pearson r vs Φ	0.363	0.831	0.883	—
Spearman ρ	—	0.936	0.822	—

Table 2: The ten-rule panel. Bold marks the diagnostic dissociations: identity (max H, zero Φ), XOR (zero bivariate TE, positive Φ — synergy blind spot), Sierpiński rule 90 (non-trivial LZ, zero Φ — self-similar over-prediction). Φ is orthogonal to H_{out} but tracks the inter-element measures (LZ, TE). Source: `companion/verify-ledger.json`.

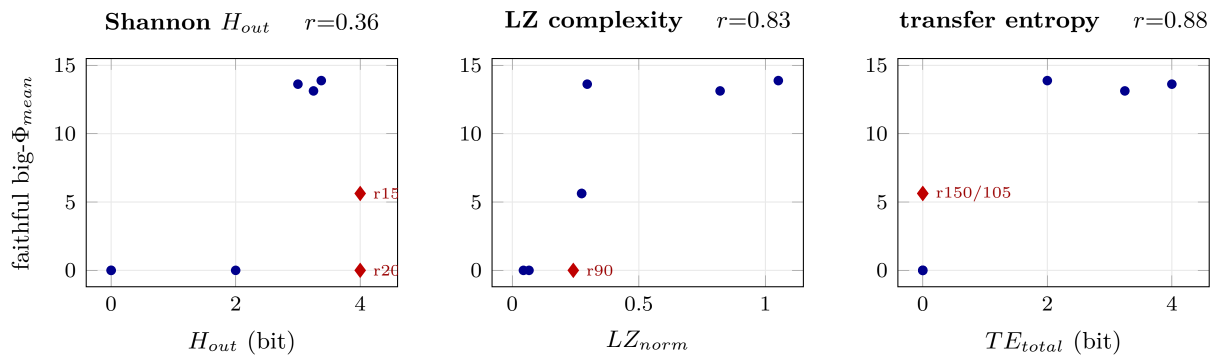

Figure 2: Faithful IIT 4.0 big-Φ vs. three classical information measures on the identical ten-rule ECA panel (blue dots; red diamonds mark the diagnostic dissociations). **Left:** Shannon output entropy — orthogonal ($r = 0.36$); at $H = 4$ bit, Φ spans 0 (identity 204) to 5.6 (XOR 150). **Centre:** LZ complexity — aligned ($r = 0.83$); rule 90 has non-trivial LZ yet $\Phi = 0$, the self-similar over-prediction. **Right:** transfer entropy — aligned ($r = 0.88$); XOR rules 150/105 sit at TE$= 0$ yet $\Phi = 5.6$, the synergy blind spot. Each classical measure has a blind spot Φ does not.

cell's own present — so it is invisible to pairwise TE conditioned only on the target's own past. The reported $r = 0.883$ is therefore an *under*-estimate; a multivariate / conditional transfer entropy would recover the XOR rules.

7.4 Structure, not density, governs Φ ( )

A complementary parity-network experiment isolates topology. At matched edge count (four edges), a scale-free hub integrates ($\Phi_{\text{mean}} = 6.81$) while a regular four-cycle does not ($\Phi = 0$); the empty graph gives 0 and the complete graph 5.6, so the ordering is non-monotone in density (the hub exceeds the denser complete graph). Edge *count* does not determine integration; edge *arrangement* (cut-resistance) does. We flag a confound honestly: the four-cycle's zero is partly a parity-on-even-cycle decoupling artifact, and a regular cycle is not a random Erdős–Rényi graph, so the literal "scale-free > random" claim is only weakly probed (Tab. 3); the robust statement is the weak form, that topology rather than density governs Φ.

Synthesis. Reading the three correlations together (Tab. 2): Φ is orthogonal to the single-system measure (Shannon entropy) and aligned with the inter-element measures (LZ, transfer

network	**edges**	Φ_{mean}	**note**
scale-free hub (paw)	4	6.81	matched density
regular cycle (4-cyc)	4	0.00	matched density
empty	0	0.00	null anchor
complete (K_4)	6	5.625	integration ceiling

Table 3: Structure experiment (parity dynamics, $N = 4$). At four edges, the hub integrates where the cycle does not; the hub also exceeds the denser complete graph. Topology, not edge count, governs Φ. The cycle's zero is partly a parity-even-cycle decoupling confound (§9).

entropy). Neither aligned measure equals Φ: LZ over-counts self-similarity, bivariate TE under-counts synergy. Φ fills both gaps. The structure experiment closes the loop — when only the wiring is changed, Φ changes from 0 to 6.8. The through-line is that Φ measures *how elements relate*, which is why no single classical scalar reproduces it, and why IIT posits it as a distinct quantity.

8 Discussion

Why entropy dissociates but flow does not. Shannon output entropy is a property of a system's input–output map considered as a single channel: a bijection is maximally informative regardless of whether its cells interact. Integration, by construction, is a property of the *joint* cause–effect structure — it asks what is lost when the system is cut. The identity rule is the sharpest illustration: it is a perfect channel ($H = N$) made of N independent perfect channels, so cutting it loses nothing ($\Phi = 0$). Transfer entropy and LZ complexity correlate with Φ precisely because they are sensitive to inter-element dependence — TE explicitly (directed conditional dependence) and LZ implicitly (a space–time diagram is incompressible only when distant cells co-determine each other). This is why the single-system measure is orthogonal and the relational measures align.

The two blind spots are complementary. LZ *over*-predicts on rule 90: a Sierpiński gasket is visually intricate and compresses poorly under a one-dimensional LZ parse, yet the rule's cells are pairwise-XOR with no irreducible whole, so $\Phi = 0$. Bivariate TE *under*-predicts on rules 150/105: XOR-of-three is the canonical synergistic gate, invisible to any pairwise conditional measure. A measure can thus mistake pattern for integration (LZ) or miss integration hidden in higher-order coupling (bivariate TE). Φ, computed over the full cause–effect structure rather than a fixed-order statistic, is subject to neither failure — which is the operational content of IIT's claim that Φ is a distinct quantity, not merely a third correlate.

Relation to partial-information decomposition. The TE blind spot is exactly the synergy term in the Williams–Beer decomposition [6]: pairwise TE captures unique and redundant information flow but not synergy. A conditional / multivariate transfer entropy (causation entropy) would recover the XOR rules and likely raise the TE–Φ correlation above the reported $r = 0.88$; our bivariate choice is the most conservative, and the fact that it already correlates strongly is the point.

Structure as the unifying axis. The topology experiment (§7.4) removes the dynamics-rule confound: holding edge count fixed and changing only the wiring moves Φ from 0 (cycle) to 6.8 (hub), and the hub exceeds the denser complete graph. Read with the measure results, the through-line is that Φ is a relational quantity — it scores how a system's parts constrain one another, which is why it tracks flow and complexity, ignores raw information, and depends on wiring rather than density.

9 Limitations and honest caveats

- **Small N.** Exact big-Φ is computed at $N = 4$; absolute r values depend on the ten-rule panel composition. The directions (orthogonal vs. aligned) are robust to panel choice; the exact coefficients are not.
- **One estimator each.** Output-image entropy, single-seed LZ76, and lag-1 *bivariate* TE are one choice each. The XOR synergy blind spot (§7.3) is specifically a property of *bivariate* TE; multivariate / causation-entropy variants would recover it. Endpoints (constant=low, bijection=max entropy) are definition-robust; mid-range magnitudes are not.
- **State-averaged Φ.** Faithful Φ is state-dependent; we average over states, so directions are trusted and absolute magnitudes are hedged.
- **Structure confound.** The cycle's $\Phi = 0$ (§7.4) is partly a parity-on-even-cycle decoupling artifact, and a cycle is not a random ER graph; the robust claim is the weak form (topology not density), and a clean ER ensemble needs $N \geq 5$ (engine cost; deferred).
- **Structure-cut, not absolute PyPhi calibration.** The engine computes a spirit-faithful structure-cut big-Φ; absolute-scale comparison to reference IIT implementations is out of scope. Our conclusions are about *correlation presence/absence* and are robust to a scale offset.
- **No phenomenal claim.** ECA rules are substrate proxies. "XOR integration," "life-themed," etc. are structural labels, not claims about phenomenal consciousness. The result is a measurement fact, not a metaphysics.

10 Future work

Four follow-ups are pre-specified. *(i) Recover the synergy term.* A conditional / multivariate transfer entropy (causation entropy) should lift rules 150/105 off the TE $= 0$ axis and is predicted to raise the TE–Φ correlation above 0.88; the size of that lift is a direct estimate of how much of Φ is synergistic on this panel. *(ii) Larger panels and N.* The exact engine is tractable to $N \approx 8$; a 256-rule sweep and an $N \geq 5$ run would turn the ten-point correlations into distributions and let the structure experiment use a genuine Erdős–Rényi ensemble, removing the parity-even-cycle confound of §7.4. *(iii) Self-reference.* A companion experiment in this lane finds that closing a self-loop creates or destroys a self-consistent fixed point depending on base topology — an "I-as-fixed-point" result that extends the relational reading of Φ to self-modelling substrates. *(iv) Beyond toy substrates.* The same three-measure triangulation can be run on small biological or artificial recurrent networks where an approximate Φ is computable, testing whether the entropy-orthogonality and the two blind spots survive outside cellular automata.

11 Reproducibility

Code: `https://github.com/dancinlab/anima` (hypotheses `H_287`–`H_290` under `HEXAD/LIFE/`, with smoke sources and `run.log` per experiment). Data: `companion/` (`verify-ledger.json` carries every panel value, Pearson/Spearman coefficient, pre-registered falsifier and its pass/fail; `pr-roll.json` the landing PRs #582–585; `session-journal.md` the build narrative). Engine: the faithful IIT 4.0 cause–effect library is reused, not reinvented (no new engine code). All experiments: deterministic, \$0, CPU-only, no GPU, no language model; re-run bit-for-bit identical. Build: `make` (xelatex $\times 3$ + bibtex). Figures: `make figures`.

Acknowledgments. Experiments were designed, run, and written up in a single foreground session with an LLM collaborator (Claude); all numeric results are deterministic outputs of the cited hexa-native smokes, independently re-runnable without the model.

A Pre-registered falsifier ledger

Every experiment froze its falsifiers before the headline values were read; each is reproduced from `companion/verify-ledger.json`. The gate columns are measurement-validity checks (anchors, bounds, determinism); the *verdict* row is the hypothesis outcome, which for H_287 is an intended falsification (the closed negative). Notably no falsifier was edited after the fact: H_289's robustness falsifier and the structure confound are reported as found.

Exp.	measure / axis	gate	hypothesis verdict
H_287	Shannon entropy	11/11	$r = 0.363 < 0.5$ (reductive falsified)
H_288	LZ complexity	9/9	$r = 0.831$, $\rho = 0.936$
H_290	transfer entropy	8/8	$r = 0.883$, $\rho = 0.822$
H_289	network topology	4/4	hub 6.81 > cycle 0 (confound-flagged)

Table 4: Per-experiment gate counts and frozen-falsifier verdicts. Engine anchors (rule 204, rule 0 $\rightarrow \Phi$=0) are reproduced to machine precision in all four; all four re-run bit-for-bit identically.

B Engine anchors

The faithful IIT 4.0 engine is validated, before any correlation is read, against two states whose Φ is hand-derivable. Rule 204 maps each cell to its own current value ($\text{next}_i = c$); the system is a direct product of N one-cell mechanisms, so the system cut is reducible and $\Phi = 0$ at every state. Rule 0 maps every state to the all-zero state; with no state-dependence there is no integrated cause and $\Phi = 0$. The engine reproduces both to machine precision. Because the panel additionally contains a max-entropy/zero-Φ rule (204), a positive-Φ/zero-TE rule (150), and a non-trivial-LZ/zero-Φ rule (90), the dissociations of §7 are anchored on independently checkable points rather than on the fitted correlation alone.

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